

1. Declare a structure named Playlist to store details of a song: title (string), artist (string), and duration in seconds (integer). Initialize one Playlist variable with your favorite song's details and print each field.

```
2. #include <stdio.h>
3.
4. // Define the structure
5. struct Playlist {
6.     char title[50];
7.     char artist[50];
8.     int duration;
9. };
10.
11. int main() {
12.     // Initialize a Playlist variable
13.     struct Playlist song = {
14.         "Perfect",
15.         "Ed Sheeran",
16.         263
17.     };
18.
19.     // Print the details
20.     printf("Song Details:\n");
21.     printf("Title      : %s\n", song.title);
22.     printf("Artist     : %s\n", song.artist);
23.     printf("Duration   : %d seconds\n", song.duration);
24.
25.     return 0;
26. }
```

2. Create a structure called FoodItem to store Zomato-style menu data: itemName (string), price (float), and rating (float). Initialize an array of 3 FoodItem variables with real menu items and display their details using a loop.

```
#include <stdio.h>

// Define the structure
struct FoodItem {
    char itemName[50];
    float price;
    float rating;
};

int main() {
    // Initialize an array of 3 FoodItem structures
    struct FoodItem menu[3] = {
        {"Paneer Butter Masala", 249.00, 4.6},
        {"Veg Biryani", 199.00, 4.4},
        {"Margherita Pizza", 299.00, 4.7}
    };
};
```

```

// Display the menu using a loop
printf("----- Zomato Menu -----\\n\\n");

for (int i = 0; i < 3; i++) {
    printf("Food Item %d\\n", i + 1);
    printf("Item Name : %s\\n", menu[i].itemName);
    printf("Price      : Rs. %.2f\\n", menu[i].price);
    printf("Rating     : %.1f\\n", menu[i].rating);
    printf("-----\\n");
}

return 0;
}

```

3. Define a nested structure called MovieShow for a BookMyShow-style app: Movie (string), Screen (integer), and a nested structure Time with hours and minutes (integers). Create and initialize a MovieShow variable for any movie and print its details in the format 'Movie: X, Screen: Y, Time: HH:MM'.

```

#include <stdio.h>

// Nested structure for Time
struct Time {
    int hours;
    int minutes;
};

// Structure for MovieShow
struct MovieShow {
    char movie[50];
    int screen;
    struct Time showTime;
};

int main() {
    // Initialize the MovieShow structure
    struct MovieShow show = {
        "Avengers: Endgame",
        5,
        {7, 30}
    };

    // Print the details
    printf("Movie: %s, Screen: %d, Time: %02d:%02d\\n",
        show.movie,
        show.screen,
        show.showTime.hours,
        show.showTime.minutes);
}

```

```
    return 0;
}
```

4. Build a structure called InstaProfile with fields: username (string), followers (integer), and a nested structure Bio with fields: description (string) and age (integer). Initialize an InstaProfile variable with your own details and display all fields.

```
#include <stdio.h>

// Nested structure for Bio
struct Bio {
    char description[100];
    int age;
};

// Structure for Instagram Profile
struct InstaProfile {
    char username[50];
    int followers;
    struct Bio bio;
};

int main() {
    // Initialize the InstaProfile structure
    struct InstaProfile profile = {
        "Ridham_Dolera",
        1500,
        {"Student and Python Developer", 20}
    };

    // Display the profile details
    printf("----- Instagram Profile -----\\n");
    printf("Username      : %s\\n", profile.username);
    printf("Followers       : %d\\n", profile.followers);
    printf("Bio             : %s\\n", profile.bio.description);
    printf("Age            : %d\\n", profile.bio.age);

    return 0;
}
```